Breast Cancer Classification using BUSI Dataset (MobileNetV2)

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1 Introduction

Breast cancer is one of the most frequently diagnosed cancers among women worldwide. Early detection significantly improves the chances of successful treatment and survival. Medical imaging techniques such as mammography, ultrasound, and MRI are commonly used for detecting abnormalities in breast tissue.

Among these techniques, ultrasound imaging is widely used because it is safe, cost-effective, and does not expose patients to radiation. However, interpreting ultrasound images manually can be time-consuming and depends heavily on the experience of radiologists.

In this project, a deep learning based classification system is developed to automatically classify breast ultrasound images into three categories:

- Benign
- Malignant
- Normal

The BUSI (Breast Ultrasound Images) dataset is used for this task, and a MobileNetV2 pretrained deep learning model is applied using transfer learning to perform the classification.

2 Dataset Description

The dataset used in this project is the BUSI (Breast Ultrasound Images) Dataset obtained from Kaggle. The dataset contains ultrasound images categorized into three classes.

2.1 Classes in Dataset

Class	Description
Benign	Non-cancerous tumor
Malignant	Cancerous tumor
Normal	No tumor present

Table 1: BUSI Dataset Classes

2.2 Number of Samples

Class	Number of Images
Benign	437
Malignant	210
Normal	133
Total	780 Images

Table 2: Number of Images per Class

The dataset also includes segmentation mask images which highlight tumor regions. However, in this project only the original ultrasound images were used for classification, and the mask images were removed during preprocessing.

3 Methodology

The overall workflow of the project consists of several steps including data preprocessing, model training, and performance evaluation.

3.1 Data Collection

The BUSI dataset was downloaded from Kaggle using the `kagglehub` library and stored in separate folders based on class labels.

3.2 Data Preprocessing

The preprocessing steps include:

- Removing mask images
- Resizing images to a fixed size (224×224)
- Converting images into tensors
- Preparing images for deep learning model input

3.3 Data Augmentation

To improve model performance and reduce overfitting, data augmentation techniques were applied:

- Horizontal flip
- Random rotation
- Image resizing and normalization

3.4 Dataset Splitting

The dataset was divided into three parts:

Dataset	Percentage
Training	70%
Validation	15%
Testing	15%

Table 3: Dataset Split

This ensures that the model is trained and evaluated properly using unseen data.

3.5 Model Selection

In this project, the MobileNetV2 pretrained model was used for image classification. Transfer learning was applied by replacing the final classification layer to classify the three classes:

- Benign
- Malignant
- Normal

MobileNetV2 is a lightweight convolutional neural network designed for efficient image classification while maintaining good performance.

3.6 Model Training

The model was trained using the following parameters:

Parameter	Value
Model	MobileNetV2
Optimizer	Adam
Learning Rate	0.0001
Loss Function	Cross Entropy Loss
Epochs	15
Batch Size	8

Table 4: Training Parameters

4 Results

4.1 Dataset Visualization

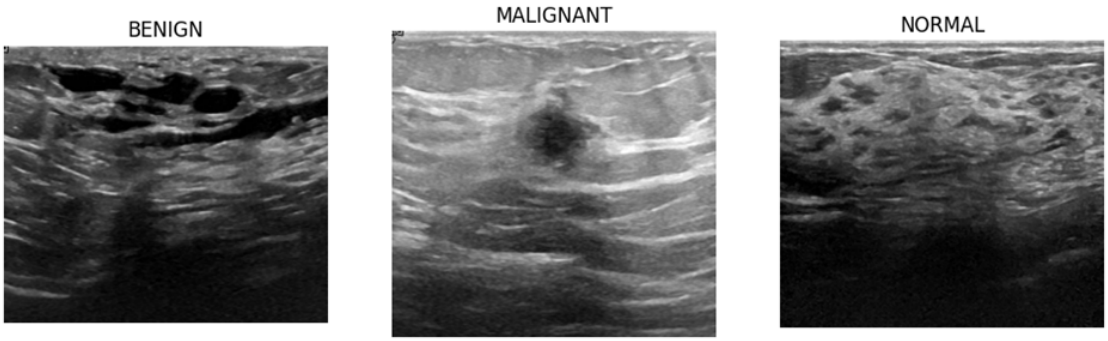

Figure 1: Sample Images from BUSI Dataset

4.2 Dataset Distribution

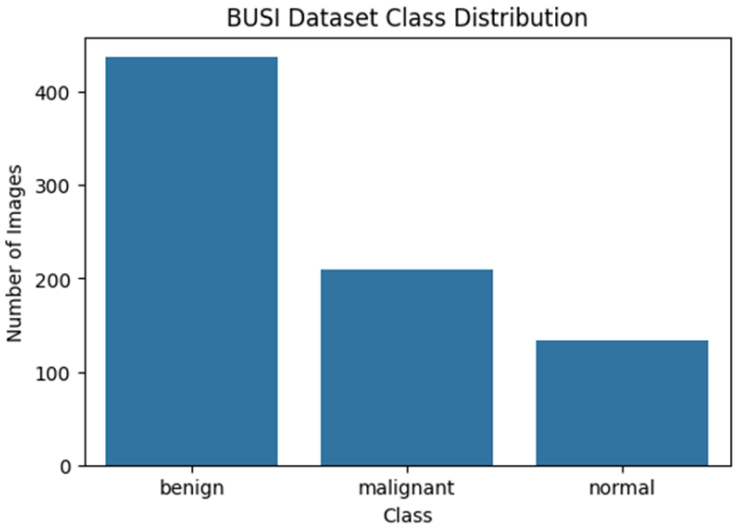

Figure 2: Distribution of Images Across Classes

4.3 Training Performance and Confusion Matrix

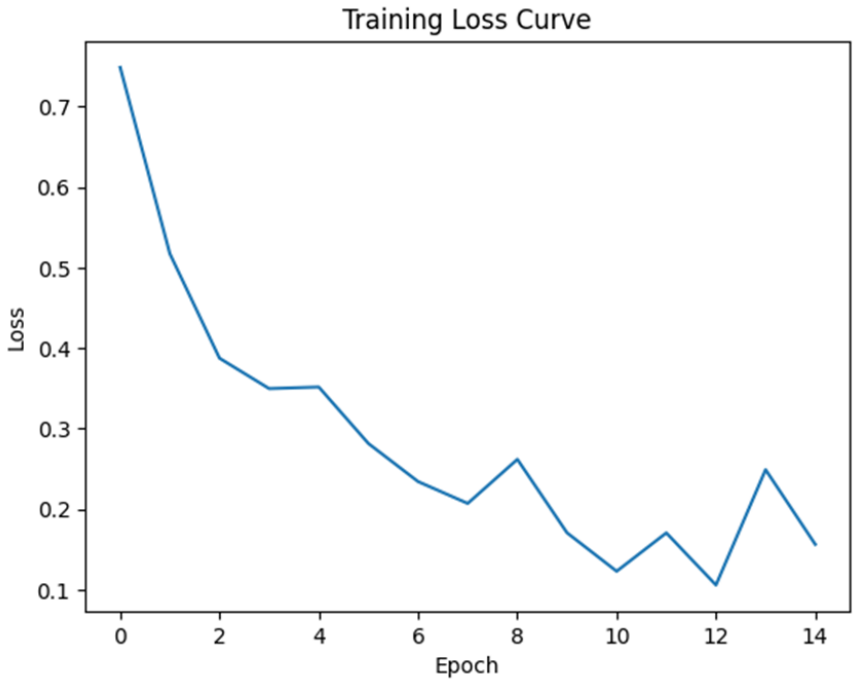

Figure 3: Training Loss Curve

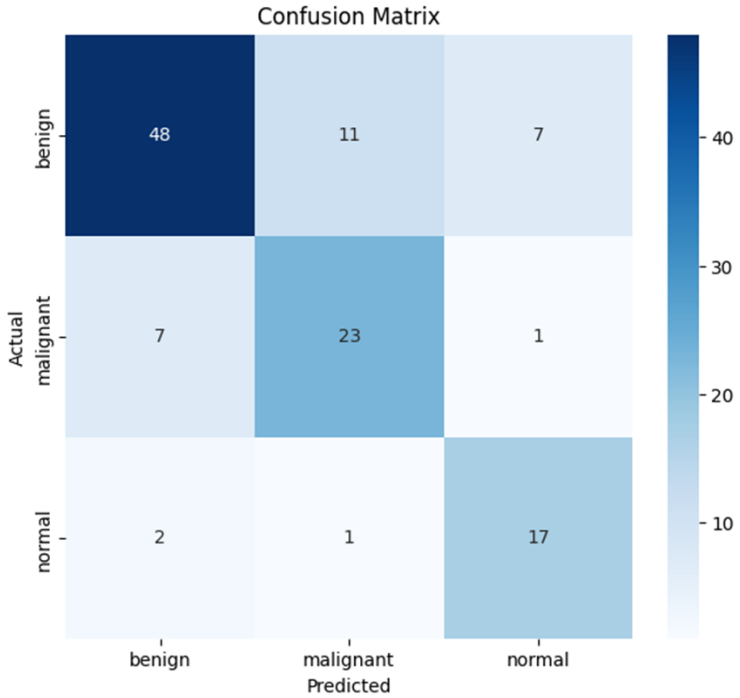

Figure 4: Confusion Matrix for Classification Results

4.4 Classification Report

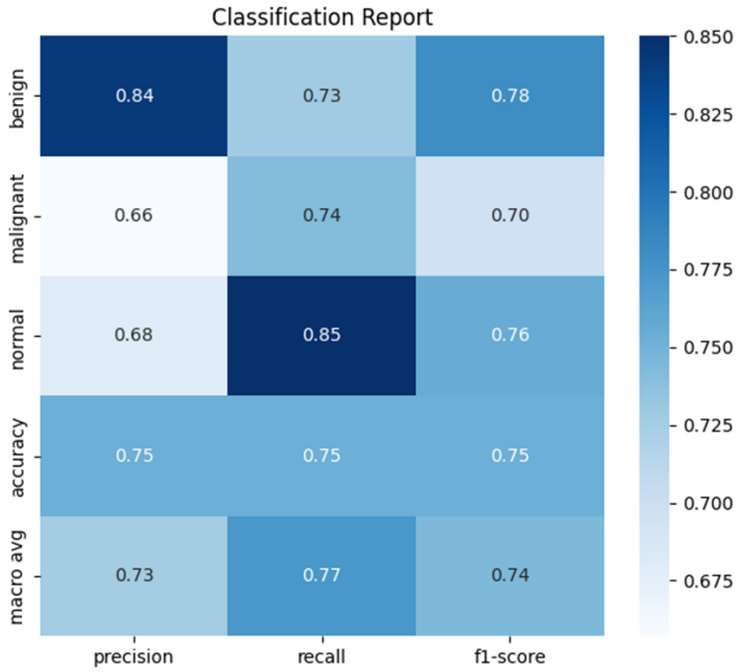

Figure 5: Classification Report Visualization

4.5 Prediction Probability Visualization

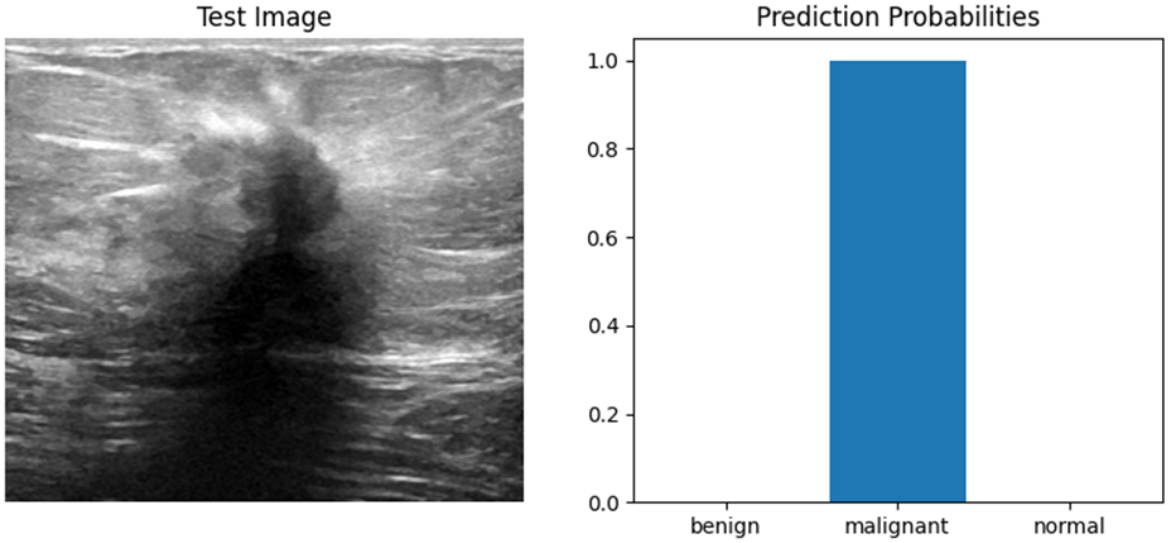

Figure 6: Prediction Probability for Sample Test Image

5 Conclusion

In this project, a deep learning model was developed to classify breast ultrasound images using the BUSI dataset.

The dataset was preprocessed and augmented to improve model performance. A MobileNetV2 pretrained convolutional neural network was applied using transfer learning to classify images into three categories: Benign, Malignant, and Normal.

The experimental results demonstrate that deep learning models can effectively classify medical ultrasound images and assist in early breast cancer detection. Such automated systems can support medical professionals by providing faster and more consistent diagnostic insights.